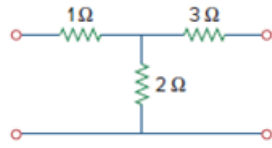


Problem

Find the transmission parameters for the circuit in Fig. 19.16 (see Practice Prob. 19.3).

Figure 19.16



Step-by-step solution

Step 1 of 9

Refer to Figure 19.16 in the text book.

Connect the voltage source V_1 as shown in Figure 1. Identify the currents and voltages in the circuit.

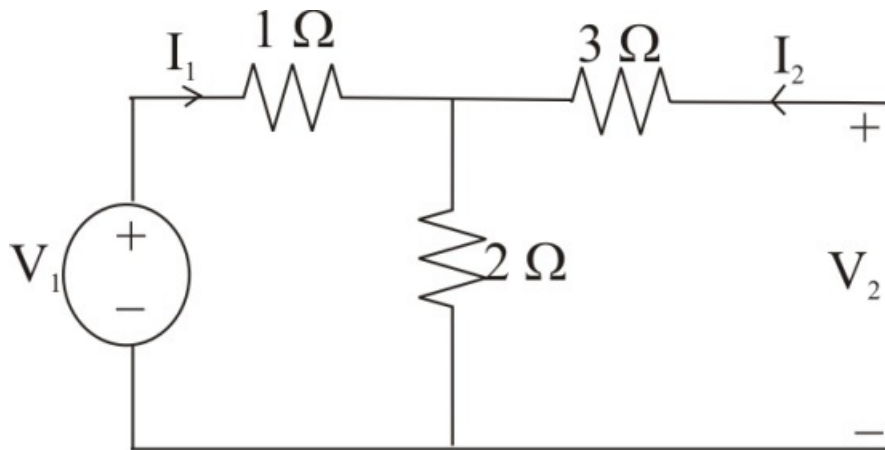


Figure 1

Step 2 of 9

Consider the expression for the voltage V_1 .

$$\begin{aligned} V_1 &= I_1(1+2) \\ &= I_1(3) \end{aligned}$$

Therefore, the expression for the voltage V_1 is $I_1(3)$.

Consider the expression for the voltage V_2 .

$$\begin{aligned} V_2 &= I_1(2) \\ &= I_1(2) \end{aligned}$$

Therefore, the expression for the voltage V_2 is $I_1(2)$.

Step 3 of 9

Determine the expression for the open circuit voltage ratio A .

$$A = \frac{V_1}{V_2}$$

Substitute $I_1(3)$ for V_1 and $I_1(2)$ for V_2 .

$$A = \frac{I_1(3)}{I_1(2)} \\ = 1.5$$

Therefore, the expression for the open circuit voltage ratio A is 1.5.

Step 4 of 9

Determine the value of the open circuit transfer admittance C .

$$C = \frac{I_1}{V_2} \text{ Substitute } I_1(2) \text{ for } V_2.$$

$$C = \frac{I_1}{I_1(2)} \\ = 500 \text{ mS}$$

Therefore, the value of the open circuit transfer admittance C is 500 mS.

Step 5 of 9

Connect the voltage source V_1 and short the port 2 terminals as shown in Figure 2. Identify the currents and voltages in the circuit.

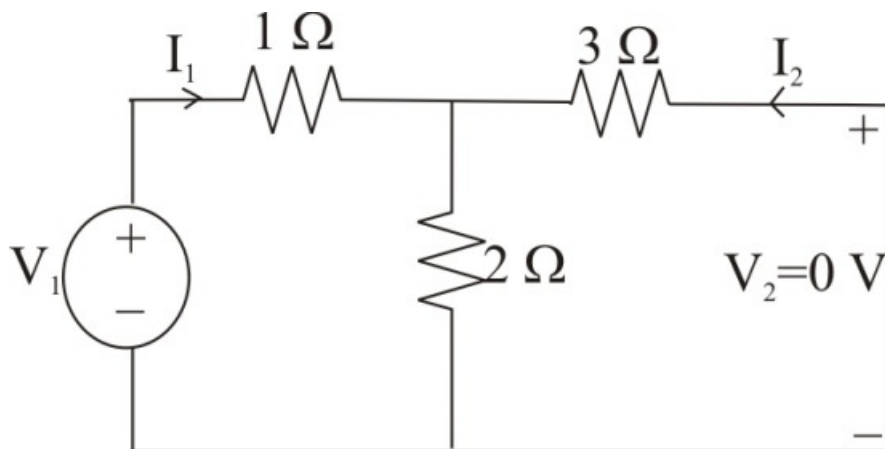


Figure 2

Step 6 of 9

Consider the expression for the voltage V_1 .

$$\begin{aligned}V_1 &= I_1 (1 + 2 \parallel 3) \\&= I_1 \left(1 + \frac{(2)(3)}{2+3} \right) \\&= I_1 \left(1 + \frac{6}{5} \right) \\&= I_1 (1 + 1.2)\end{aligned}$$

$$V_1 = I_1 (2.2)$$

Therefore, the expression for the voltage V_1 is $I_1 (2.2)$.

Step 7 of 9

Consider the expression for the current I_2 .

$$\begin{aligned}I_2 &= -I_1 \left(\frac{2}{2+3} \right) \\&= -I_1 \left(\frac{2}{5} \right) \\&= -I_1 (0.4)\end{aligned}$$

Therefore, the expression for the current I_2 is $-I_1 (0.4)$.

Step 8 of 9

Determine the expression for the negative short circuit transfer impedance B .

$$B = -\frac{V_1}{I_2}$$

Substitute $I_1 (0.4)$ for I_2 and $I_1 (2.2)$ for V_1 .

$$\begin{aligned}B &= -\frac{I_1 (2.2)}{I_1 (0.4)} \\&= -5.5 \, \Omega\end{aligned}$$

Therefore, the expression for the negative short circuit transfer impedance B is $\boxed{-5.5 \, \Omega}$.

Step 9 of 9

Determine the value of the negative short circuit current gain D .

$$D = -\frac{I_1}{I_2}$$

Substitute $-I_1 (0.4)$ for I_2

$$\begin{aligned}D &= -\frac{I_1}{I_1 (0.4)} \\&= -25\end{aligned}$$

Therefore, the value of the negative short circuit current gain D is $\boxed{25}$.